Supplementary Material

Learning to Monitor the Reliability Horizon of Imagined Futures

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This supplement documents the environments and data collection, the world models and monitor, the statistical procedure, the censoring analysis, additional results including the size-matched control and the ensemble-size ablation, the two configurations rejected as invalid instruments, the planning experiment, and the executable verification of the information boundary.

Pre-registration documents, the deviation ledger, and per-seed statistics accompany the code. Every number reported in the main paper and in this supplement is emitted programmatically from stored result artifacts by `scripts/make_paper_tables.py` and `scripts/make_paper_figures.py`; neither script recomputes an experiment.

1 Experimental Details

1.1 Environments and data collection

All configurations are MiniGrid environments [Chevalier-Boisvert et al., 2023] with `agent_view_size` $= 5$, giving a $5 \times 5 \times 3$ egocentric symbolic observation, and `max_steps` $= 400$. Because the view is egocentric, grid size never enters the observation and one world model applies to every configuration by construction.

	base environment	perturbation	role
A	`DoorKey-16x16`	—	training distribution
B	`FourRooms`	—	unseen layout topology
C	`DoorKey-16x16`	action slip 0.15	stochastic dynamics
D	`MultiRoom-N6`	—	long corridors
E	`SimpleCrossingS11N5`	—	unseen maze family
F	`DoorKey-16x16`	obs. noise 0.15	rejected instrument
F2	`DoorKey-16x16`	obs. noise 0.70	rejected instrument
F3	`DoorKey-16x16`	action dropout 0.20	accepted instrument

Action slip replaces the requested action by a uniformly random one. *Observation noise* corrupts a fraction of object-channel cells in the egocentric view, leaving dynamics untouched. *Action dropout* replaces the requested action by MiniGrid's `done` action, a verified no-op that does not terminate the episode, so the agent's intent silently fails to execute.

Collection policy. Trajectories are collected by a forward-biased explorer ($p(\text{forward}) = 0.6$) that additionally pursues scripted key/door subgoals with probability 0.35, for 120 steps per episode. An earlier BFS-based explorer was discarded during pre-registration because it produced $P(\text{moved} \mid \text{forward}) = 0.985$: shortest paths never walk into walls, which removes the epistemic uncertainty the experiment measures. Under the forward-biased policy that quantity is 0.48–0.67 across configurations.

Splits. Layout seeds are disjoint by construction: world-model training $[0, 3000)$, monitor training $[3000, 4200)$, validation $[4200, 4800)$, test $[4800, 5400)$, and shifted configurations $[100000, 100600)$. The monitor trains on episodes the world model never saw, so H^* is not inflated by memorisation. For the transfer experiments each shifted configuration is further partitioned by episode id ($\bmod 5 \in \{0, 1, 2\}$ train, 3 validation, 4 test), with the test portion evaluation-only in every regime.

1.2 World model

Encoder: per-cell embeddings of object ($11 \to 16$), colour ($6 \to 6$) and state ($3 \to 6$), flattened over the 5×5 view and concatenated with a one-hot direction and a carrying flag, then a two-layer MLP to 384 units. Recurrent core: a GRU cell [Cho et al., 2014] with deterministic state 512 and Gaussian stochastic state 48, with prior $p(z_t \mid h_t)$ and posterior $q(z_t \mid h_t, e_t)$. Decoder heads on $[h_t, z_t]$ predict dir (4-way), moved, carrying and door_open. Approximately 2.85M parameters.

Training deliberately mixes filtering and open-loop imagination: for each sequence a split point $c \sim \mathcal{U}\{5, \ldots, 20\}$ is drawn, the posterior is used for $t < c$ and the prior alone for $t \geq c$, so the model is optimised for the open-loop rollout that is later evaluated. Loss is a weighted cross-entropy over the heads (weights 1.0 for dir, 2.0 for moved, 0.3 for the remaining two) plus a free-nats KL term. Optimisation: AdamW, learning rate 3×10^{-4}, weight decay 10^{-5}, batch 64, sequence length 50, 6000 steps, gradient-norm clipping at 100, mixed precision on GPU. Ensembles use $K = 5$ members differing in seed and data order.

Transformer variant. The Transformer world model replaces the recurrence with a causal self-attention stack [Vaswani et al., 2017] ($d_{\text{model}} = 256$, 2 layers, 4 heads, GELU, pre-norm). Observations are replaced by a learned mask token wherever unavailable, so training and imagination are single parallel passes and there is no autoregressive latent feedback. It exposes the same interface and head set. One definitional difference is unavoidable: its one-step context error uses a fully masked pass, because a per-token "everything strictly before t" mask is not expressible in one forward pass. This withholds *more* information than the GRU version, so it cannot flatter the Transformer.

1.3 Monitor

g_ϕ is an MLP with two hidden layers of 256 units and SiLU activations, producing logits over $H^* \in \{0, \ldots, 30\}$, trained with cross-entropy for 60 epochs (AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch 512). Inputs are standardised using statistics from the training split only and clipped to ± 10 s.d. The point estimate is the mean of the predictive distribution, and the same convention is applied to every baseline including isotonic regression, so comparisons are like-for-like. Checkpoint selection uses validation MAE; test data informs no choice.

Feature groups. *latent*: member-0 belief state $[h_t, z_t]$. *apriori*: one-step-ahead entropy and ensemble Jensen–Shannon divergence, prior/posterior standard deviations and their KL at the anchor, one-step tracking error over the observed context, and distance from the anchor latent to the nearest training latent. *imag*: per-horizon ensemble disagreement, entropy, prior standard deviation and spread of imagined displacement at $h \in \{1, 2, 5, 10, 20, 30\}$ plus summaries, and the horizon at which imagined rollouts first diverge by one cell (the online-truncation baseline). *actions*: counts and prefix sums of the planned action sequence. *hand*: object-type counts in the current view, visible free cells ahead, unseen cells ahead, carrying and door flags, direction. Variants used in the paper are `full` (all groups), `no_latent`, `uncertainty_only` (*apriori*+*imag*), `strict_apriori` (all except *imag*), `hand_actions_only` and `actions_only`. In the cross-architecture experiment only `no_latent` is used, because its column *names* are identical across architectures while total feature dimensionality is not (634, 346, 362); the code asserts name equality before any cross-architecture comparison.

1.4 Statistics

Confidence intervals use a bootstrap that resamples whole *episodes*, not anchors, because anchors within an episode are strongly dependent [Efron and Tibshirani, 1993]. We use 2000 resamples and report 95% percentile intervals. Comparisons between methods are paired: the same resampled episode set is used for both methods and the interval is computed on the difference. Reported statistics are MAE, RMSE, Spearman ρ, Pearson r, Kendall τ and R^2; the main text reports MAE, ρ and R^2. Calibration uses the expected calibration error of $P(H^* \geq h)$ in ten equal-width probability bins, discarding bins with fewer than 20 states.

The Claim-A ceiling uses isotonic regression [Barlow et al., 1972] with the direction chosen automatically, fitted on the evaluation split. This is deliberately optimistic: it is the minimum of (**??**) attainable by any monotone map, with no generalisation gap.

1.5 Censoring

H^* is right-censored at $H_{\max} = 30$. On A-test 20.7–21.7% of anchors are at the ceiling across seeds. Consequences: MAE is inflated relative to an uncensored target; the upper tail of H^* is unidentified; and R^2 is computed against a censored variance. All methods including baselines and the ceiling face the identical censoring, so comparisons *between* methods remain valid, but absolute values should not be read as estimates of an uncensored horizon.

1.6 Compute

The main run trained 15 world models (3 seeds $\times$ $K = 5$) in 4.4 hours on a single Tesla P100. The cross-architecture study added 9 models. All follow-up experiments (H5, H6, H8, H9) reuse stored features or checkpoints and require no world-model retraining. Total: 24 world models. Runs were executed on Kaggle with PyTorch 2.5.1+cu121; the preinstalled build did not support the allocated P100's compute capability, so a Pascal-compatible build is installed at kernel start.

2 Additional Results

2.1 Full hypothesis ledger

Table **??** in the main text lists all nine hypotheses. Two procedural notes belong with it.

First, in two places our scoring code and our written decision rule diverged. In H8 the pre-registration assigned NOT SUPPORTED only when the first prediction failed and PARTIAL when the first two held, leaving the case "first holds, second fails" unspecified; the code defaulted to NOT SUPPORTED and we report that rather than choosing a label after seeing the outcome. In H9 the pre-registration states that the verdict is evaluated only over configurations passing the validity gate, which would make two predictions uninformative, while the code scored one of them as a failure. Both paths reach the same verdict, so nothing changes, but we record the discrepancy.

Second, the ensemble-size ablation (seed 0) shows the effect does not require an ensemble:

K	monitor MAE ↓	ρ ↑	ceiling MAE
1	6.908	0.533	8.897
3	6.582	0.547	7.809
5	6.504	0.559	7.583

At $K = 1$ there is no ensemble disagreement at all, only single-model entropy and latent statistics, and the monitor still beats its ceiling — by the largest margin of the three. Ensembling improves absolute accuracy and improves the ceiling faster than it improves the monitor.

2.2 Size-matched transfer control

Table 1 repeats the transfer matrix with every regime subsampled to the A-only training-anchor count (22,200), so all regimes see equal data.

training regime	test A	test B	test C	test D
A	*+0.580*	+0.218	+0.213	+0.135
A+B	*+0.575*	*+0.747*	+0.278	+0.759
A+C	*+0.573*	+0.411	*+0.477*	+0.463
A+D	*+0.575*	+0.694	+0.287	*+0.844*
A+B+C	*+0.569*	*+0.745*	*+0.466*	+0.762

Table 1: Size-matched control for Table **??**. Spearman ρ, mean over three seeds; italic entries are configurations seen in training.

The critical cell A+B+C →D gives MAE 3.622 and $\rho = 0.762$, against 3.584 and 0.766 unmatched. The effect is therefore not explained by sample count alone. We do not claim that diversity is its only cause: severity of the donor shift covaries with its presence, and we did not manipulate the two independently.

2.3 Rejected instruments

Two configurations failed the pre-registered validity gate (a new configuration counts as an instrument only if H^* is at least 40% below A's *and* the no-exposure monitor transfers to it with $\rho < 0.35$):

	perturbation	H^*	no-exposure ρ
F	obs. noise 0.15	15.23	+0.472
F2	obs. noise 0.70	12.72	+0.359
F3	action dropout 0.20	3.67	+0.238

Observation noise never bites: even at 70% cell corruption H^* falls only from 15.5 to 12.7. An action-conditioned model with sixteen steps of belief context does not depend heavily on the corrupted channel. F and F2 are reported as invalid instruments rather than as evidence about transfer; predictions targeting them are uninformative by rule. The severity ladder used to select F2 and F3 is in `results/aleatoric/calibration.json`.

A useful control follows from this. The regime A+F2 — exposure to a configuration that barely degrades the model — teaches nothing transferable: $\rho = 0.226$ on B and 0.087 on D, at or below the no-exposure baseline. Exposure helps only when the donor genuinely degrades the world model.

2.4 Planning experiment (weak test)

The adaptive-planning arm sets the imagination horizon of a random-shooting model-predictive controller to the monitor's prediction, clipped to $[1, 30]$, and is charged for the probe rollout used to produce that prediction ($K \cdot H_{\max}$ imagined states per decision). Candidate action sequences are drawn from the same forward-biased distribution used to collect training data. Targets require a detour: shortest-path distance must exceed Manhattan distance. Sixty layouts, paired across arms.

arm	success	imagined states / decision	mean H
fixed H=3	0.917 ± 0.036	288	3.0
fixed H=5	0.933 ± 0.032	480	5.0
fixed H=10	0.767 ± 0.055	960	10.0
fixed H=20	0.517 ± 0.065	1920	20.0
fixed H=30	0.367 ± 0.062	2880	30.0
adaptive	0.517 ± 0.065	2637	25.9

Success falls monotonically with horizon, so the compute–performance frontier has no interior optimum and an adaptive rule has nothing to find. The detour manipulation did not take: mean detour was 0.13 cells, with 57 of 60 layouts falling through to the least restrictive tier. We therefore report this as a weak test of downstream utility rather than a refutation. The adaptive arm's specific failure — selecting $H \approx 26$ where 3–5 is optimal — is the anti-conservative bias of §**??** reappearing under the state distribution that model-predictive control itself induces.

2.5 Information-boundary verification

The monitor may use no observation, state, or error at time $> t$. This is verified executably rather than asserted: every field of the evaluation window strictly after the anchor is replaced with random values, the features are recomputed, and all feature columns must be bit-identical while the target must change. Corrupting the future leaves the features unchanged and moves mean H^* from 6.45 to 0.01. The planned action sequence is deliberately preserved by the corruption, since a planner genuinely knows it.

3 Deviations from Pre-Registration

Ten deviations are logged in `DEVIATIONS.md` with their timing relative to test evaluation. Six occurred before any test data was examined:

D1 Configurations changed after a geometry survey that used no neural network: the original set left the agent travelling ≈ 2.7 cells in 30 steps while seeing 4 cells ahead, so the planned path never left the visible region and H^* was degenerate.

D2 Collection policy changed from a BFS explorer to a forward-biased one, for the reason given in §1.

D3 Splits restructured to give the monitor its own world-model-held-out split.

D4 Pilot test seeds moved outside the main test range after an early diagnostic reported aggregate accuracies on an overlapping range; that diagnostic informed no threshold and its models and data were discarded.

D5 Two analysis additions: the privileged-state comparison, and selection of the monitor variant on validation MAE.

D6 Planning task made non-myopic, and the adaptive arm charged for its probe rollout.

Four are post-main-run and scoped to follow-ups: D7 records that all reported numbers come from Kaggle and that a briefly started local run was terminated without producing any result; D8 records the planning re-run and its residual limitation; D9 records that the H5 protocol was frozen blind; D10 records the size-matched control and the new 60/20/20 partitions, and notes that consequently H6 numbers are not directly comparable to full-split numbers elsewhere.

References

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